

Definition

- **Data:** Collection of **raw facts**.
- **Data structure** is representation of the logical relationship existing between individual elements of data.
- **Data structure** is a specialized format for organizing and storing data in memory that considers not only the elements stored but also their relationship to each other.

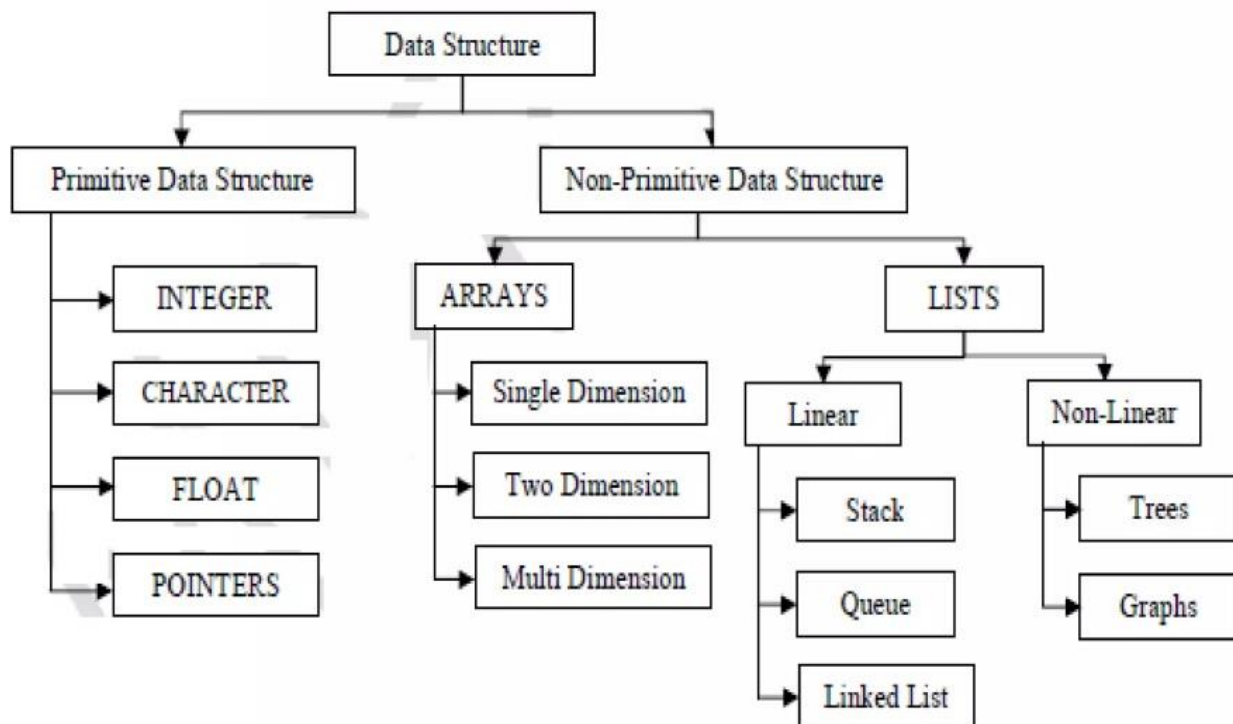
Introduction

- Data structure affects the design of both structural & functional aspects of a program.
Program=algorithm + Data Structure
- You know that a algorithm is a step by step procedure to solve a particular function.

Classification of Data Structure

- Data structure are normally divided into two broad categories:
 - Primitive Data Structure
 - Non-Primitive Data Structure

Classification of Data Structure



Primitive Data Structure

- There are basic structures and directly operated upon by the machine instructions.
- *Data structures that are directly operated upon the machine-level instructions are known as primitive data structures.*
- Integer, Floating-point number, Character constants, string constants, pointers etc, fall in this category.

Primitive Data Structure

- The most commonly used operation on data structure are broadly categorized into following types:
 - Create
 - Selection
 - Updating
 - Destroy or Delete

Non-Primitive Data Structure

- There are more sophisticated data structures.
- *The Data structures that are derived from the primitive data structures are called Non-primitive data structure.*
- The non-primitive data structures emphasize on structuring of a group of homogeneous (same type) or heterogeneous (different type) data items.

Non-Primitive Data Structure

Linear Data structures:

- *Linear Data structures are kind of data structure that has homogeneous elements.*
- The data structure in which elements are in a sequence and form a liner series.
- Linear data structures are very easy to implement, since the memory of the computer is also organized in a linear fashion.
- Some commonly used linear data structures are **Stack, Queue and Linked Lists.**

Non-Linear Data structures:

- *A Non-Linear Data structures is a data structure in which data item is connected to several other data items.*
- Non-Linear data structure may exhibit either a hierarchical relationship or parent child relationship.
- The data elements are not arranged in a sequential structure.
- The different non-linear data structures are **trees and graphs.**

Non-Primitive Data Structure

- The most commonly used operation on data structure are broadly categorized into following types:
 - Traversal
 - Insertion
 - Selection
 - Searching
 - Sorting
 - Merging
 - Destroy or Delete

Different between them

- **A primitive data structure** is generally a basic structure that is usually built into the language, such as an integer, a float.
- **A non-primitive data structure** is built out of primitive data structures linked together in meaningful ways, such as a or a linked-list, binary search tree, AVL Tree, graph etc.

Description of various Data Structures : Arrays

- An array is defined as a set of finite number of homogeneous elements or same data items.
- It means an array can contain one type of data only, either all integer, all float-point number or all character.

One dimensional array:

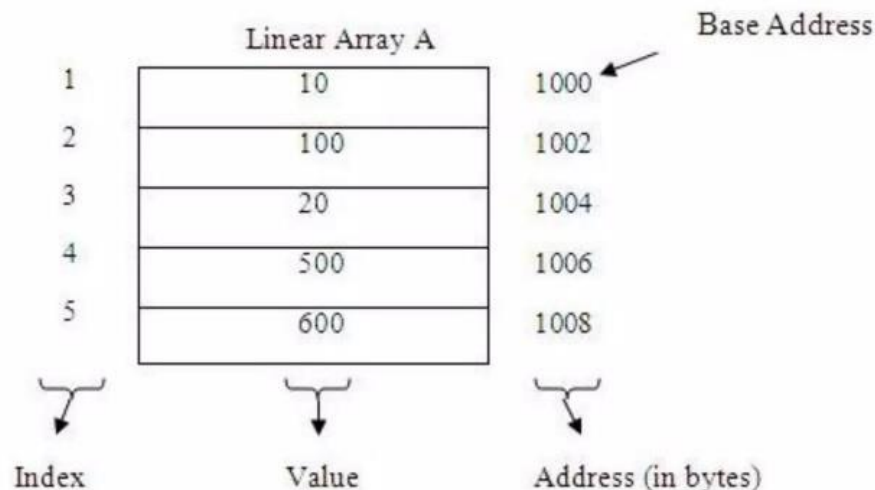
- *An array with only one row or column is called one-dimensional array.*
- It is finite collection of n number of elements of same type such that:
 - can be referred by indexing.
 - The syntax Elements are stored in continuous locations.
 - Elements x to define one-dimensional array is:
- **Syntax: Datatype Array_Name [Size];**
- Where,
Datatype : Type of value it can store (Example: int, char, float)
Array_Name: To identify the array.
- Size : The maximum number of elements that the array can hold.

Arrays

- Simply, declaration of array is as follows:
`int arr[10]`
- Where int specifies the data type or type of elements arrays stores.
- “arr” is the name of array & the number specified inside the square brackets is the number of elements an array can store, this is also called sized or length of array.

Represent a Linear Array in memory

- The elements of linear array are stored in consecutive memory locations. It is shown below:



Arrays

- The elements of array will always be stored in the consecutive (continues) memory location.
- The number of elements that can be stored in an array, that is the size of array or its length is given by the following equation:
$$(\text{Upperbound} - \text{lowerbound}) + 1$$
- For the above array it would be $(9 - 0) + 1 = 10$, where 0 is the lower bound of array and 9 is the upper bound of array.

Arrays types

- Single Dimension Array
 - Array with one subscript
- Two Dimension Array
 - Array with two subscripts (Rows and Column)

Basic operations of Arrays

- Some common operation performed on array are:
 - Traversing
 - Searching
 - Insertion
 - Deletion
 - Sorting
 - Merging